Large-Scale Monitoring of Soil Organic Matter and Salinization Using Satellite Hyperspectral Imagery and a Deep Residual Attention Network

Chao Niu, Libin Zhu, Kun Tan, *Senior Member, IEEE*, Xue Wang, Run Niu, and Xiaojun Xu

***Abstract*—Soil degradation threatens environmental health and sustainable land use, with declining soil organic matter (SOM) and increasing salinization being two key manifestations. Accurate assessment of the spatial distribution of SOM and salinity is of great importance for both agricultural productivity and ecological conservation. This article proposes a novel end-to-end deep-learning model—the residual attention regression network (RARNet)—to enable quantitative estimation of soil parameters based on satellite-borne hyperspectral imagery. RARNet combines 2-D spectral convolution, residual feature learning, and attention mechanisms to jointly capture local spectral patterns and global spectral dependencies, thereby improving spatial–spectral feature representation. The experimental results obtained over a 17 000 km^2 black-soil region using a satellite hyperspectral dataset show the improved predictive performance of RARNet. Under the fivefold cross-validation setting, the validation coefficients of determination (R^2) are 0.7291 and 0.8368, the root-mean-square error (RMSE) values are 2.0986 and 2.4806, and the mean absolute error (MAE) values are 1.7449 and 2.0152 for SOM and soil salinity, respectively. Compared with representative deep learning baselines, including long short-term memory (LSTM) and convolutional neural network (CNN) with spectral attention, RARNet achieved higher average performance. It also improved R^2 by approximately 10%–15% compared with the ResNet backbone and by 15%–25% compared with the CARS-XGBoost method. The trained model was applied to the study area, generating large-scale spatial distribution maps of SOM and salinity. The mapping results revealed that the region suffers from low SOM levels and widespread salinization, with over half the area showing a salt content of 1.0 g/kg or higher. Overall, the proposed RARNet provides a feasible framework for large-scale mapping of SOM, salinization, and their spatial patterns in the study area. This demonstrates the potential of combining satellite hyperspectral imagery with deep learning for the monitoring of soil properties and land degradation. Although further validation is required for other soil types and climates, the current findings provide a quantitative reference for land restoration and regional ecological risk assessments.**

***Index Terms*—Attention mechanism, regression network, satellite hyperspectral, soil organic matter (SOM), soil salinization.**

I. Introduction

SOIL degradation poses significant challenges to environmental health and sustainable development [1]. Global estimates indicate that approximately 1 billion hectares of land are affected by salinization, representing about 7% of the Earth's land surface [2], [3]. This issue is particularly acute in the black-soil region of Northeast China. As one of the most fertile agricultural areas and a vital commercial grain base globally, it has suffered severe degradation due to long-term intensive land utilization. This degradation is primarily manifested as a decline in topsoil thickness and a reduction in soil organic matter (SOM) content [4]. The soil organic carbon (SOC) content in this region has decreased by 3.07 g/kg over the past 35 years [5]. Furthermore, soil salinization has become increasingly prominent. The region currently contains approximately 16.93 million hectares of salinized or alkalinized zonal soils [6], posing a significant threat to agricultural productivity and the ecological environment [7]. SOM and soil salinization are coupled with each other. Biologically, elevated salinity inhibits plant growth, drastically reducing fresh organic carbon inputs [8]. Physiochemically, high salinity disperses soil aggregates, exposing SOM to microbial attack and significantly accelerating carbon mineralization [9]. Real-time and accurate monitoring of SOM and salinity is essential for ensuring sustainable agricultural production and ecological conservation. Traditional soil quality assessment relies on extensive field sampling, which is both costly and time-consuming, hindering large-scale mapping efforts [10]. Remote sensing technology enables large-scale monitoring and the extraction of multidimensional information, providing a viable alternative for soil quality assessment. However, multispectral remote sensing is constrained by the limited spectral resolution, making it challenging to establish the relationships between soil properties and spectral information. Hyperspectral remote sensing, with its superior spectral diagnostic capabilities, has emerged as a powerful tool for accurate soil monitoring. In

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particular, airborne hyperspectral data, which are characterized by a high spatial and spectral resolution, are particularly valuable for precision soil quality monitoring [11], [12]. Recent advancements in satellite hyperspectral technology, marked by the deployment of full-spectrum satellites such as GaoFen-5, ZY-1, PRISMA, and ENMAP, have further expanded the potential for large-scale soil quality monitoring [13], [14]. However, hyperspectral images are characterized by high dimensionality and strong inter-band correlations, and they are often affected by sensor noise, atmospheric absorption, and low-signal-to-noise bands, which pose challenges for data processing and analysis, potentially impeding the effective application of these data [15], [16], [17]. Therefore, it is necessary to develop efficient feature extraction techniques and robust regression models to accurately estimate soil parameters.

The conventional workflow for hyperspectral soil parameter estimation relies on a two-step process: feature analysis and regression modeling. The feature analysis aims to extract relevant spectral features and reduce the data dimensionality using methods such as the Pearson correlation coefficient (PCC), principal component analysis (PCA), or variable importance in projection (VIP). Subsequently, machine learning algorithms such as random forest (RF), support vector machines (SVM), and artificial neural networks (ANNs) are employed to build a regression model from these extracted features [18], [19]. While this approach can achieve a reasonable predictive accuracy, its fundamental limitation lies in the separation of feature extraction and regression. These nonend-to-end modeling methods can lead to the selection of suboptimal features and the propagation of errors, ultimately hindering the model's full predictive potential.

Deep-learning-based regression networks transform the original spectral inputs into high-level feature representations through hierarchical extraction, and their strong capability for modeling nonlinear relationships offers an effective method for the regression prediction of high-dimensional spectral data [20]. This end-to-end model enables the direct prediction of soil parameters, streamlining the modeling workflow and presenting potential for accurate soil mapping [21]. In recent years, deep learning algorithms—particularly convolutional neural networks (CNNs), recurrent neural networks (RNNs), long short-term memory (LSTM) networks, and Transformers—have been successfully applied to the estimation of soil parameters. Based on traditional CNN architectures, Tang et al. [22] have adopted spectral-to-image transform techniques to convert multisource spectra into dual-band images, which are subsequently processed by a multichannel CNN to enhance the performance of SOC estimation. Jain et al. [23] proposed the TLM-Stack framework—a novel ensemble architecture combining TabNet, LSTM, and multilayer perceptron (MLP)—to effectively estimate diverse soil nutrients. Dong et al. [24] developed the attention-based LSTM (A-LSTM) model, which incorporates a scaled dot-product attention mechanism to dynamically adjust time-step weights, focusing on critical spectral sequences to enhance predictive robustness. Sangwan et al. [25] proposed the Transformer-CNN framework, which couples CNN and Transformer branches to simultaneously capture local spectral absorptions and long-range dependencies, utilizing adaptive fusion to improve the robust estimation of multiple soil properties. In addition, introducing a residual module into a deep network can avoid vanishing and exploding gradients, improve the generalization performance of the model, and further enhance the ability of deep features to represent soil parameters [26]. However, these models are primarily designed for point-based spectral modeling. When applied to hyperspectral imagery, they fail to leverage the spatial contextual information and the inherent spatial autocorrelation of soil properties. Constrained by the relatively limited spatial resolution of satellite hyperspectral imagery, this neglect of spatial-spectral coupling makes it difficult for existing architectures to accurately capture the intricate nonlinear interactions between soil parameters and spectral responses. Moreover, the conventional point-based deep regression models exhibit a limited capacity to explicitly model the key spectral bands and spectral responses associated with soil parameters such as salinity and organic matter. These models also cannot dynamically reweight the features based on the variations in spatial–spectral characteristics across different regions, thereby constraining both the feature representation efficiency and generalization performance. Consequently, their performance remains dependent on manually selected features and may be unstable in heterogeneous satellite hyperspectral scenes. This limitation motivates the development of a novel modeling strategy that can automatically learn informative spectral responses related to SOM and salinity from satellite hyperspectral data, while accounting for the characteristics of high spectral resolution, strong interband correlation, and relatively low spatial resolution.

To address these limitations, it is necessary to develop a deep learning architecture with spectral structural representation capability, which is capable of jointly learning both shared and task-specific feature representations from raw hyperspectral inputs. In this article, we propose the residual attention regression network (RARNet), a novel regression architecture that integrates a deep residual network with an attention mechanism. Different from existing transformer-based or hybrid models, RARNet reshapes the 1-D spectral sequence into a pseudo-2-D spectral representation to facilitate convolution-based learning of local cross-band interactions. This representation does not imply real spatial adjacency among spectral bands. By combining the pseudo-2-D spectral representation, residual learning, and attention-based feature recalibration, RARNet improves the representation of informative spectral features. This model directly extracts the relevant information for soil parameter estimation from the original spectra, streamlining the feature extraction, selection, and analysis steps, and thereby improving the accuracy of soil parameter retrieval. Accordingly, this study aims to: 1) develop RARNet for the end-to-end estimation of SOM and salinity from satellite-borne hyperspectral imagery; 2) evaluate its performance against representative baseline models, including traditional machine-learning methods and deep-learning architectures; and 3) apply the trained model to generate large-scale SOM and salinity maps for the study area, providing preliminary spatial information for soil degradation assessment and future soil management.

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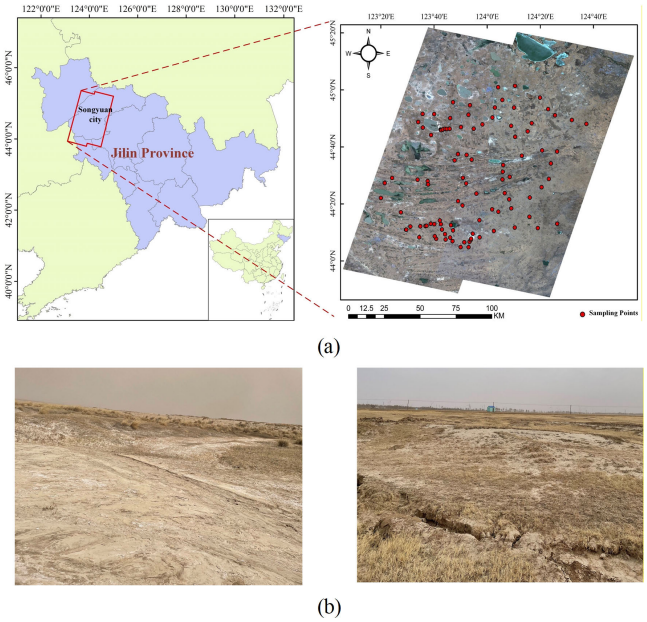


Fig. 1. Location of the study area. (a) Geographic location of the study area and the sampling points. (b) Field photographs of the saline-alkali soil area.

II. Materials

A. Study Area

The study area is situated in the city of Songyuan, Jilin Province, Northeast China, specifically in the southern reaches of the Songnen Plain. The geographical coordinates of the study area span from 43°59′N to 45°32′N latitude and 123°6′E to 126°11′E longitude, encompassing an area of approximately 17 000 square kilometers. Fig. 1 shows the location of the study area and the spatial distribution of the field sampling sites. A spatial representation of the study area and the distribution of the sampling points is depicted in Fig. 1(a).

The dominant soil types within the study area are chernozem soil (also known as black soil), meadow soil, and saline-alkali soil. The saline-alkali soil is classified as soda saline-alkali soil, which is characterized by high salinity and alkalinity. This area represents one of the three major concentrated distribution zones of soda saline-alkali soils globally [27]. Field photographs of the saline-alkali soil are presented in Fig. 1(b).

B. Soil Sampling and Laboratory Analysis

Soil sampling was conducted during the bare soil period from April to May, after the majority of the soils had been tilled and the cultivated land surface was free of snow and relatively dry. Soil samples were collected in mid-to-late April 2023, with a total of 100 samples obtained. The principle underlying the selection of sampling points was to ensure that each point was surrounded by large expanses of bare soil, thereby mitigating the interference from non-bare soil pixels. Constrained by the brief bare-soil window and quasi-synchronous satellite overpasses, the 100 sampling sites were distributed across the major soil types and degradation levels to improve the representativeness of the dataset for model development. A five-point sampling method was selected for soil collection. At each sampling point, five topsoil subsamples at a depth of 0–20 cm were collected from the center and four corners of a 1×1 m grid. These subsamples were thoroughly mixed to obtain a standard composite sample for laboratory analysis.

TABLE I
Statistical Analysis of the SOM and Soil Salinity Content in the Study Area

PARAMETER	SOM	Salinity
Min (g/kg)	4.30	0.46
Max (g/kg)	24.30	47.20
Mean (g/kg)	13.68	2.82
STD (g/kg)	4.20	6.80
CV (%)	30.70	241.13

The SOM content was determined using the potassium dichromate oxidation method with external heating [28]. This method involves oxidation of the SOC in the soil samples with an excess of $K_2Cr_2O_7$ in a concentrated H_2SO_4 solution. The remaining $K_2Cr_2O_7$ is then back-titrated with a standard ferrous sulfate solution. The amount of SOC is subsequently calculated based on the consumption of ferrous sulfate. Finally, to convert the oxidized SOC into SOM content, a standard correction factor of 1.724 was applied, following the Chinese Forestry Industry Standard (LY/T 1237-1999) [29].

The measurement of soil salinity content was conducted using a gravimetric method, following the Chinese Forestry Industry Standard (LY/T 1251-1999) [30]. Soluble salts were extracted from the soil using a 5:1 water-to-soil ratio. The soil–water mixture was subjected to shaking and centrifugation to separate the supernatant. The supernatant was then filtered to obtain the extract. The extract was evaporated to dryness using a water bath, followed by oven drying to remove any residual organic matter. The mass of the remaining residue, representing the soluble salts, was then determined by weighing [30].

The results of the SOM and soil salinity content determination are summarized in Table I. The SOM content in the study area ranges from 4.3 to 24.3 g/kg, with a coefficient of variation (CV) of 30.7%. This indicates a moderate degree of variability and a relatively stable data distribution. The normality of the SOM content data was assessed using the Shapiro–Wilk test. The resulting p-value is 0.118 (Statistic $W = 0.9793$), which is greater than 0.05, suggesting that the SOM content data follow a normal distribution. The salinity content ranges from 0.46 to 47.2 g/kg, with a mean of 2.82 g/kg. The CV is 241.13%, indicating strong spatial heterogeneity and a highly skewed distribution caused by localized severe salinization. The Shapiro–Wilk test for salinity yielded a p-value < 0.001 (Statistic $W = 0.3212$), statistically confirming a nonnormal distribution, which is highly characteristic of natural, localized salt accumulation [31], [32].

Furthermore, Global Moran's I was calculated using an inverse-distance spatial weight matrix to quantify the spatial

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TABLE II
PARAMETERS OF THE ZY1-02 E AHSI

PARAMETER	ZY1-02E AHSI
Bands	166 (VNIR: 76, SWIR: 90)
Spectral range	400–2500 nm
Spatial resolution	≤30m
Spectral resolution	VNIR: 10 nm, SWIR: 20 nm
Accuracy of the absolute radiometric calibration	<7%
Accuracy of the relative radiometric calibration	<3%
Swath width	≥60 km
Revisit cycle	3 days
Orbital altitude	778 km
Launch date	2021-12-26

autocorrelation among the sampling points. The Moran's I value for SOM was 0.477 (Z-score = 5.857, $p < 0.001$), and for salinity was 0.713 (Z-score = 10.308, $p < 0.001$). Both results indicate highly significant positive spatial autocorrelation, particularly for salinity, which exhibits strong spatial clustering.

C. Hyperspectral Data Acquisition

The satellite hyperspectral imagery was acquired using the advanced hyperspectral imager (AHSI) mounted on the ZY1-02E satellite. The imager parameters are listed in Table II. A total of six hyperspectral images were obtained on March 29, 2023, and April 21, 2023, covering approximately 17 000 square kilometers. Both acquisition dates had clear skies, and the illumination and atmospheric conditions were relatively consistent, with an aerosol optical depth of approximately 0.13 at 550 nm and a water vapor content of approximately 1.6 g/cm^2. In addition, land-cover types showed minimal changes between the two dates, making the data suitable for this experiment. The images were subjected to radiometric calibration, atmospheric correction using the fast line-of-sight atmospheric analysis of spectral hypercubes (FLAASH) module to retrieve accurate surface reflectance, removal of water-vapor absorption bands and low-signal-to-noise bands around 1350–1430 nm and 1800–1930 nm, and image mosaicking to generate the final data for the study area, resulting in 147 valid bands for subsequent model input. The radiometric consistency of the preprocessed images was assessed using the overlapping regions between adjacent scenes, and the mean relative difference of the surface reflectance was less than 5% [33]. Soil pixels were extracted using an SVM classifier with a radial basis function (RBF) kernel. Training samples were manually delineated by cross-referencing the high-resolution imagery. The model achieved an overall accuracy of 94.6%, and the soil classification results are illustrated in Fig. 2. The extracted soil pixels are mainly distributed in cultivated land and exposed bare-soil areas, while water bodies, vegetation-covered areas, and built-up land are effectively excluded. This soil mask was used to constrain subsequent SOM and salinity inversion. All remaining valid hyperspectral bands were used as model inputs.

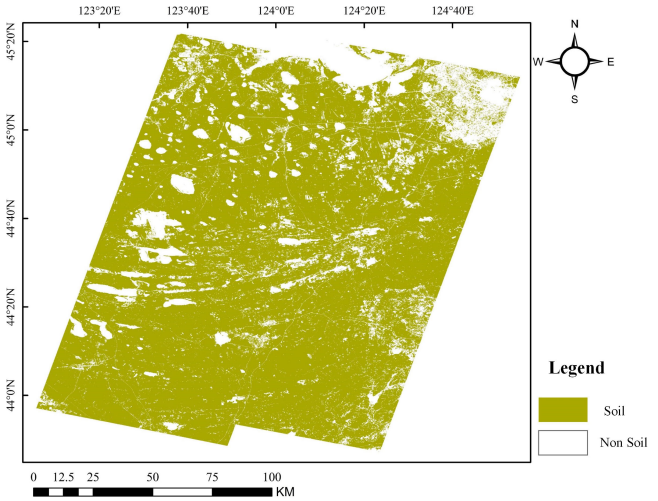


Fig. 2. Soil classification results for the study area.

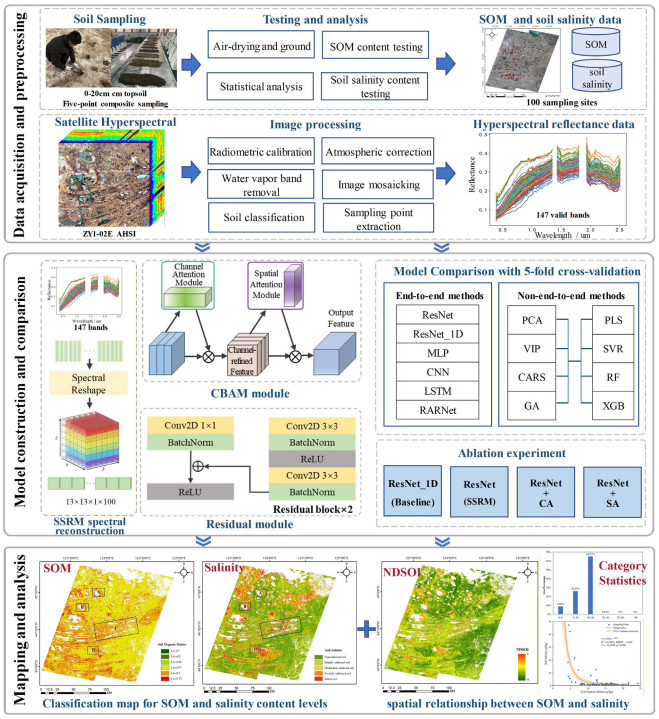


Fig. 3. Schematic flow of the study.

III. METHOD

In this study, we developed a novel deep regression network by integrating laboratory-measured SOM and salinity data with satellite hyperspectral imagery. The aim was to generate spatial distribution maps of the SOM and salinity content in the study area. The schematic flow of the study is shown in Fig. 3. The workflow consists of three stages: data preprocessing, model construction, and mapping analysis. First, field samples and satellite hyperspectral images were processed to build paired spectral–soil parameter datasets. Second, RARNet was developed using a spectral structural reconstruction module (SSRM), residual feature extraction, and CBAM-based feature recalibration, and it was compared with representative baseline

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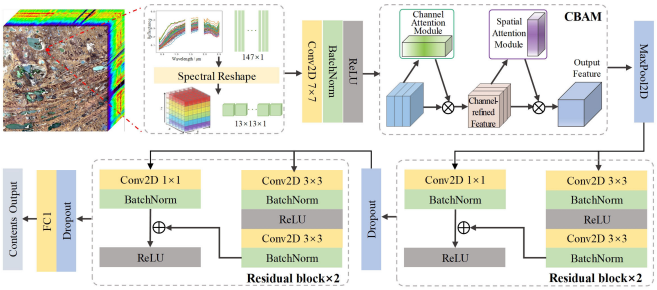


Fig. 4. Structure of the RARNet network.

models. Finally, the trained model was used to generate SOM and salinity maps and conduct level classification and NDSOI-based spatial analysis.

A. Residual Attention Regression Network

Based on the ResNet backbone architecture, a novel deep-learning model—the RARNet—was developed. The overall network architecture is shown in Fig. 4. The input data consist of hyperspectral reflectance spectra with 147 bands, which are preprocessed through atmospheric correction and the removal of water-vapor absorption bands. In order to effectively capture the local correlations between adjacent hyperspectral bands, an SSRM is introduced. This module reconstructs the original 147-D spectral vector into a 13×13 2-D tensor via zero padding. This transformation does not imply a real spatial structure among spectral bands. Instead, it provides a structured representation that enables 2-D convolution kernels to capture local cross-band interactions.

A 7×7 convolution is first applied to extract the feature maps, which are stacked along the channel dimension to form a high-dimensional tensor. Batch normalization (BN) and rectified linear unit (ReLU) activation follow. To enhance the feature representation, a convolutional block attention module (CBAM) is integrated, comprising a channel attention (CA) module with a 7×7 kernel and a spatial attention (SA) module with a 1×1 kernel. The output features are then processed by a max-pooling layer and fed into a residual module containing four identical residual blocks. Each block performs 2-D convolution, BN, and ReLU, with residual connections formed by adding the block input—after 1×1 convolution and BN—to the output. Dropout layers are added after the second and fourth blocks to mitigate overfitting. Finally, a fully connected layer maps the extracted features to the predicted soil parameter values. The mean-squared error (mse) loss was utilized to optimize the model.

B. Model Comparison

To evaluate the performance of the proposed RARNet model in estimating SOM and salinity from hyperspectral data, an ablation study was conducted, and several end-to-end deep learning regression methods were employed for comparison. Nonend-to-end methods using feature analysis and regression modeling were also included in the comparative analysis. A brief introduction to the comparative methods used in this study is provided below.

1) End-to-End Methods: ResNet and ResNet_1D: The proposed RARNet method is based on the ResNet architecture. Both variants share the same core network structure and hyperparameters as the proposed RARNet without the CBAM. In particular, ResNet_1D employs 1-D convolutional layers directly on the raw spectral sequence, without utilizing the SSRM. On the contrary, ResNet first incorporates the SSRM to reshape the 1-D spectral vector into a 2-D tensor, followed by standard 2-D convolutions for feature extraction.

MLP: An MLP typically consists of multiple fully connected layers, where each neuron in a layer is connected to all the neurons in the preceding layer, without any skip connections. The MLP architecture used in this study comprised five linear layers and five BN layers, with the ReLU serving as the activation function.

LSTM: LSTM networks utilize standard gated units to capture long-range temporal dependencies while avoiding the vanishing gradient problem. In this study, the LSTM model was configured with two stacked LSTM hidden layers containing 16 neurons each. To mitigate overfitting and enhance nonlinear representation, a dropout layer and a ReLU-activated fully connected layer were added before the final regression output.

CNN: A CNN incorporates layers such as convolutional, pooling, and fully connected layers. CNNs effectively extract local features from the input data and reduce the feature dimensionality through convolution and pooling operations, subsequently performing classification or regression tasks via the fully connected layers. The CNN model employed in this research consisted of five convolutional layers, interleaved with pooling layers, and two fully connected layers, utilizing ReLU as the activation function. Furthermore, a squeeze-and-excitation (SE) module was integrated into the network to function as a spectral attention mechanism, enabling the model to dynamically recalibrate and emphasize informative hyperspectral bands.

2) Non-End-to-End Methods: To further evaluate and compare the performance of end-to-end deep feature extraction methods in soil parameter estimation, a total of 16 nonend-to-end method combinations were implemented, incorporating four feature analysis methods and four machine learning regression models. The feature analysis methods included: 1) PCA; 2) VIP; 3) competitive adaptive reweighted sampling (CARS); and 4) a genetic algorithm (GA). The four regression models were partial least squares regression (PLSR), support vector regression (SVR), RF, and extreme gradient boosting (XGBoost).

C. Model Evaluation

In this study, we employed three indices to evaluate the model performance: the coefficient of determination (R^2), the root-mean-square error (RMSE), and the mean absolute error (MAE). R^2 quantifies the goodness of fit between the predicted and true values. Its value ranges from 0 to 1, with values closer to 1 indicating a higher prediction accuracy. RMSE measures the magnitude of the errors between the predicted and actual values. A lower RMSE signifies a stronger predictive capability of the model. MAE represents the average of the

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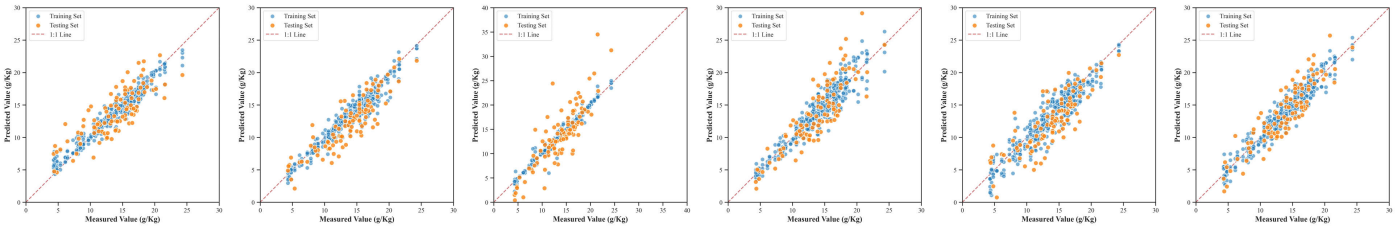


Fig. 5. Scatter plots of the measured and estimated values for SOM. (a) ResNet. (b) ResNet_1D. (c) MLP. (d) CNN. (e) LSTM. (f) RARNet.

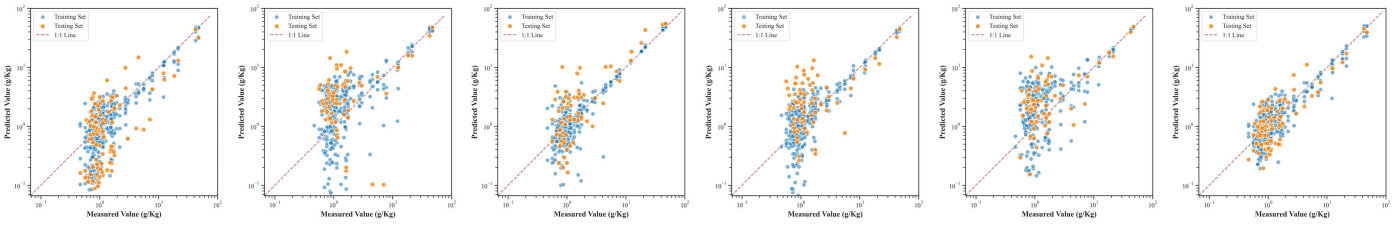


Fig. 6. Scatter plots of the measured and estimated values for salinity. (a) ResNet. (b) ResNet_1D. (c) MLP. (d) CNN. (e) LSTM. (f) RARNet.

TABLE III
MODELING ACCURACY FOR SOIL PARAMETERS OBTAINED USING THE END-TO-END METHODS

Parameter	Model	Training set			Validation set			p-value
		R^2	RMSE	MAE	R^2	RMSE	MAE	
SOM	ResNet	0.9470 ± 0.0412	0.8492 ± 0.4414	0.6671 ± 0.3499	0.6428 ± 0.0473	2.4096 ± 0.3403	2.0399 ± 0.3055	0.073
	ResNet_1D	0.9513 ± 0.0468	0.7442 ± 0.5348	0.6130 ± 0.4466	0.6055 ± 0.0591	2.5299 ± 0.3591	2.0265 ± 0.2584	0.047*
	MLP	0.9778 ± 0.0299	0.4569 ± 0.4215	0.3015 ± 0.2743	0.3425 ± 0.0749	3.2722 ± 0.4615	2.3606 ± 0.3443	0.008**
	CNN	0.8850 ± 0.0446	1.3754 ± 0.2886	1.0834 ± 0.2202	0.5829 ± 0.0512	2.6051 ± 0.3675	2.0580 ± 0.2212	0.042*
	LSTM	0.8650 ± 0.0583	1.4830 ± 0.3480	1.1337 ± 0.3000	0.5430 ± 0.0670	2.7234 ± 0.3862	2.2658 ± 0.3714	0.015**
	RARNet	**0.9120 ± 0.0309**	**1.2075 ± 0.2291**	**0.9782 ± 0.1885**	**0.7291 ± 0.0355**	**2.0986 ± 0.2963**	**1.7449 ± 0.2329**	**Ref**
Salinity	ResNet	0.9323 ± 0.0174	1.7359 ± 0.2937	1.4018 ± 0.2381	0.6596 ± 0.0938	3.4151 ± 2.3025	2.8925 ± 1.9333	0.017*
	ResNet_1D	0.8761 ± 0.0293	2.3511 ± 0.4185	1.8618 ± 0.3126	0.5247 ± 0.0367	3.9235 ± 2.2793	3.3512 ± 1.9517	0.015*
	MLP	0.9429 ± 0.0105	1.6121 ± 0.3123	1.2833 ± 0.2565	0.5810 ± 0.0518	3.6584 ± 2.1017	2.5249 ± 1.4637	0.003**
	CNN	0.9343 ± 0.0117	1.7302 ± 0.3243	1.3631 ± 0.2494	0.7019 ± 0.0345	3.0894 ± 1.7781	2.6252 ± 1.6167	0.067
	LSTM	0.8476 ± 0.0327	2.6114 ± 0.4442	2.0461 ± 0.3119	0.6430 ± 0.0410	3.3814 ± 1.9466	2.7991 ± 1.6005	0.008**
	RARNet	**0.9230 ± 0.0096**	**1.8631 ± 0.2594**	**1.5190 ± 0.2098**	**0.8368 ± 0.0316**	**2.4806 ± 1.5072**	**2.0152 ± 1.2327**	**Ref**

*The results with the highest accuracy have been highlighted in bold. Data are expressed as the mean value ± standard deviation from cross-validation. The p-values indicate the results of the Wilcoxon signed-rank test comparing each model against the proposed RARNet (Reference).

absolute differences between the predicted and actual values. A smaller MAE indicates better prediction performance for the model

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y}_i)^2} \quad (1)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{N}} \quad (2)$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i| \quad (3)$$

where $\hat{y}_i$ is the predicted value for the ith sample, y_i is the actual value for the ith sample, and N represents the total number of samples.

IV. RESULTS

A. End-to-End Methods

The dataset used for the modeling consisted of 100 sets of soil parameter laboratory measurements and their corresponding spectra. Six end-to-end methods were employed to model and estimate two soil parameters—SOM and salinity—separately. The dataset was evaluated using a fivefold cross-validation strategy. In each fold, 80% of the samples were used for training, and the remaining 20% were used for validation. Furthermore, to ascertain statistical significance, a Wilcoxon signed-rank test was performed to compare model predictions. The modeling results are presented in Table III, and the scatter plots of the measured and estimated values are shown in Figs. 5 and 6.

The fivefold cross-validation results show that RARNet achieves the highest average predictive accuracy for both SOM (R^2 = 0.7291, RMSE = 2.0986) and soil salinity

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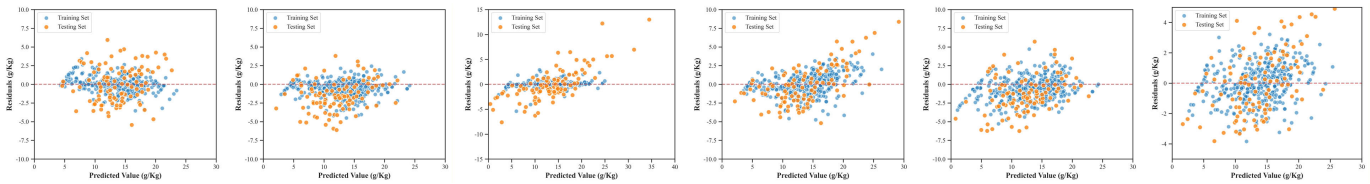


Fig. 7. Residual distribution plots for SOM. (a) ResNet. (b) ResNet_1D. (c) MLP. (d) CNN. (e) LSTM. (f) RARNet.

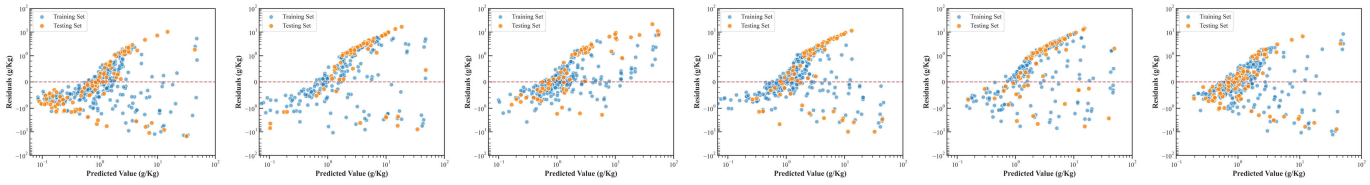


Fig. 8. Residual distribution plots for salinity. (a) ResNet. (b) ResNet_1D. (c) MLP. (d) CNN. (e) LSTM. (f) RARNet.

(R^2 = 0.8368, RMSE = 2.4806). This demonstrates that the proposed SSRM and attention mechanisms provide a more effective framework for capturing relevant spectral interactions compared to standard baselines within the constraints of this small-sample dataset. The difference in prediction performance between SOM and soil salinity may be partly related to their different statistical distributions, with soil salinity exhibiting much stronger spatial heterogeneity than SOM. Although the validation RMSE for soil salinity is relatively large compared to the dataset's mean value, this apparent discrepancy is primarily attributed to the skewed data distribution within the study area, where the maximum value reaches up to 47.20 g/kg. In this context, the RMSE accounts for a relatively small proportion of the total dynamic range, indicating that RARNet can characterize the overall spatial variation and relative differences in salt accumulation. Consequently, the model demonstrates practical utility for regional-scale mapping and spatial screening within the boundaries of the current study dataset.

Comparing ResNet and ResNet_1D, ResNet outperforms its 1-D variant, suggesting that converting input data for 2-D convolution can improve the performance. This can be attributed to 2-D convolution's enhanced capability to capture both local and global spectral features, as well as its effectiveness in suppressing noise through neighborhood information, which 1-D convolution lacks. Due to the limited sample size, the MLP tends to memorize the training noise rather than learning generalizable spectral representations. The MLP exhibited severe overfitting—achieving high training accuracy but low accuracy during validation. Compared with RARNet, the CNN model achieves a lower accuracy, highlighting the advantage of SSRM and residual connections in improving model stability and accuracy. The LSTM model shows a limited predictive accuracy for both SOM and salinity. This underperformance indicates that capturing spectral sequential dependencies is less effective than the strategy of explicitly extracting local cross-band interactions in a 2-D space.

To verify whether the differences between RARNet and the competing models were statistically significant, a Wilcoxon signed-rank test was conducted. As presented in Table III, RARNet showed statistically significant improvements over most comparison models under the current cross-validation setting. Although the differences with ResNet for SOM and the CNN for salinity are only marginally significant, RARNet still obtains the highest performance. Furthermore, a detailed diagnosis of the model errors was conducted using residual distribution plots shown in Figs. 7 and 8. The MLP model exhibits a clear discrepancy between its tightly clustered training residuals and more dispersed validation residuals. The CNN and LSTM demonstrate wider residual scattering, particularly at the extreme high and low values of SOM and salinity. In comparison, the residuals of RARNet were more concentrated around the zero-error reference line, indicating lower prediction bias and greater error stability.

B. Non-End-to-End Methods

Hyperparameter tuning for the machine learning regression models was performed using a grid search. The final modeling results are summarized in Table IV.

As shown in the table, models such as RF and XGB exhibit extreme discrepancies between their training and validation performance, which provides clear evidence of overfitting, confirming that standard machine learning algorithms are highly susceptible to the limited sample size of the current dataset. For SOM, the combination of feature analysis and machine learning regression does not yield a satisfactory prediction performance. The CARS-XGB model ranks highest on the validation set, while its R^2 value reaches only 0.5134 and its RMSE is 2.8146. In contrast, the modeling performance for soil salinity is generally better than for SOM. All four feature selection methods combined with machine learning regression produce validation set R^2 values above 0.6. Among the different comparison methods, the CARS-XGB model achieves the highest accuracy, with a validation set R^2 of 0.7416 and an RMSE of 3.7218. However, this still represents a limited accuracy when compared with the RARNet model.

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TABLE IV
MODELING ACCURACY FOR SOIL PARAMETERS OBTAINED USING THE NON-END-TO-END METHODS

Soil Parameter	Feature Analysis Method	Regression Model	Training set			Validation set		
			R^2	RMSE	MAE	R^2	RMSE	MAE
SOM	PCA	PLS	0.7183 ± 0.0386	2.4137 ± 0.1182	1.9364 ± 0.0871	0.1834 ± 0.1124	3.9142 ± 0.3128	3.1257 ± 0.2463
		SVR	0.7241 ± 0.0347	2.2618 ± 0.1047	1.7842 ± 0.0763	0.3421 ± 0.0946	3.3165 ± 0.2641	2.6348 ± 0.2114
		RF	0.9967 ± 0.0031	0.2184 ± 0.0143	0.1637 ± 0.0116	0.2874 ± 0.1285	3.4619 ± 0.3284	2.7561 ± 0.2583
		XGB	0.8652 ± 0.0268	1.5238 ± 0.0864	1.2147 ± 0.0643	0.3846 ± 0.1068	3.1482 ± 0.2536	2.4938 ± 0.2147
	VIP	PLS	0.6628 ± 0.0416	2.6143 ± 0.1348	2.1136 ± 0.0984	0.2638 ± 0.0975	3.7641 ± 0.2947	2.9862 ± 0.2318
		SVR	0.9914 ± 0.0047	0.1873 ± 0.0154	0.1462 ± 0.0121	0.1537 ± 0.1086	3.8126 ± 0.3541	3.0184 ± 0.2763
		RF	0.9986 ± 0.0020	0.0984 ± 0.0083	0.0741 ± 0.0062	0.2218 ± 0.1214	3.7143 ± 0.3418	2.9461 ± 0.2674
		XGB	0.8973 ± 0.0234	1.3418 ± 0.0764	1.0634 ± 0.0573	0.3641 ± 0.0918	3.2184 ± 0.2613	2.5642 ± 0.2184
	CARS	PLS	0.7538 ± 0.0325	2.2146 ± 0.1083	1.7634 ± 0.0821	0.4426 ± 0.0987	2.9847 ± 0.2341	2.3618 ± 0.1947
		SVR	0.6724 ± 0.0398	2.5183 ± 0.1246	2.0165 ± 0.0942	0.4218 ± 0.0894	3.1162 ± 0.2418	2.4837 ± 0.2016
		RF	0.9995 ± 0.0011	0.0461 ± 0.0042	0.0348 ± 0.0031	0.3526 ± 0.1117	3.3184 ± 0.2816	2.6418 ± 0.2284
		XGB	**0.9037 ± 0.0178**	**1.3164 ± 0.0718**	**1.0347 ± 0.0562**	**0.5134 ± 0.0796**	**2.8146 ± 0.2163**	**2.2461 ± 0.1834**
	GA	PLS	0.6631 ± 0.0362	2.5238 ± 0.1362	2.0184 ± 0.0963	0.2837 ± 0.0924	3.5186 ± 0.2842	2.7841 ± 0.2216
		SVR	0.6752 ± 0.0413	2.4361 ± 0.1284	1.9427 ± 0.0913	0.3448 ± 0.0961	3.3142 ± 0.2584	2.6318 ± 0.2164
		RF	0.9954 ± 0.0041	0.2841 ± 0.0247	0.2183 ± 0.0184	0.4136 ± 0.1056	3.1264 ± 0.2618	2.4816 ± 0.2084
		XGB	0.8732 ± 0.0285	1.5284 ± 0.0918	1.2163 ± 0.0762	0.4817 ± 0.0834	2.9183 ± 0.2314	2.3164 ± 0.1927
Soil salinity	PCA	PLS	0.8134 ± 0.0301	3.1462 ± 0.1584	2.5183 ± 0.1164	0.5132 ± 0.1056	5.4183 ± 0.4632	4.3165 ± 0.3814
		SVR	0.8326 ± 0.0287	2.8417 ± 0.1426	2.2634 ± 0.1083	0.6784 ± 0.0913	4.1863 ± 0.3418	3.3427 ± 0.2816
		RF	0.7748 ± 0.0348	3.2418 ± 0.1634	2.5841 ± 0.1248	0.7136 ± 0.0837	3.8642 ± 0.3184	3.0847 ± 0.2561
		XGB	0.9352 ± 0.0164	1.8146 ± 0.0983	1.4382 ± 0.0746	0.7164 ± 0.0935	3.8418 ± 0.3241	3.0615 ± 0.2634
	VIP	PLS	0.6138 ± 0.0438	4.2184 ± 0.2183	3.3647 ± 0.1684	0.4253 ± 0.1167	5.8164 ± 0.4826	4.6381 ± 0.3942
		SVR	0.7752 ± 0.0334	3.3461 ± 0.1682	2.6683 ± 0.1264	0.5348 ± 0.1245	5.1384 ± 0.4283	4.1162 ± 0.3418
		RF	0.7846 ± 0.0307	3.1284 ± 0.1537	2.4836 ± 0.1184	0.5837 ± 0.1053	4.8615 ± 0.3916	3.8824 ± 0.3147
		XGB	0.8835 ± 0.0231	2.4168 ± 0.1264	1.9284 ± 0.0963	0.6152 ± 0.0986	4.5832 ± 0.3826	3.6618 ± 0.2984
	CARS	PLS	0.7738 ± 0.0321	3.2186 ± 0.1764	2.5641 ± 0.1348	0.7248 ± 0.0854	3.8247 ± 0.3142	3.0481 ± 0.2516
		SVR	0.7554 ± 0.0352	3.3142 ± 0.1834	2.6418 ± 0.1362	0.6035 ± 0.0978	4.7186 ± 0.3927	3.7642 ± 0.3148
		RF	0.9856 ± 0.0065	0.8164 ± 0.0583	0.6482 ± 0.0461	0.5148 ± 0.1096	5.2163 ± 0.4582	4.1648 ± 0.3684
		XGB	**0.9338 ± 0.0157**	**1.8462 ± 0.0864**	**1.4683 ± 0.0642**	**0.7416 ± 0.0715**	**3.7218 ± 0.3014**	**2.9642 ± 0.2418**
	GA	PLS	0.9254 ± 0.0163	1.9184 ± 0.0942	1.5263 ± 0.0764	0.6738 ± 0.0927	4.2618 ± 0.3524	3.3846 ± 0.2761
		SVR	0.8653 ± 0.0248	2.5186 ± 0.1384	2.0134 ± 0.0982	0.6842 ± 0.0865	4.1638 ± 0.3461	3.3247 ± 0.2748
		RF	0.9738 ± 0.0087	1.1642 ± 0.0763	0.9284 ± 0.0582	0.7256 ± 0.0964	3.8647 ± 0.3216	3.0862 ± 0.2583
		XGB	0.9154 ± 0.0184	2.0163 ± 0.1084	1.6128 ± 0.0843	0.6457 ± 0.0992	4.4286 ± 0.3718	3.5342 ± 0.2964

*The results with the highest accuracy have been highlighted in bold. Data are expressed as the mean value ± standard deviation from cross-validation.

C. Ablation Study

To evaluate the effectiveness of the SSRM and the CBAM attention mechanism in improving the model performance, an ablation study was conducted to evaluate network architectures with different attention mechanism modules on both the SOM and salinity datasets. All architectures in this study share the identical residual network backbone, ensuring that the performance gains are attributable to the introduced modules rather than the residual connections themselves. The network architectures were progressively evaluated using the fivefold cross-validation strategy.

1) *ResNet_1D* (*Baseline*)*:* Represents a 1-D convolutional network applied directly to the raw spectral sequence. It strictly excludes both the SSRM and CBAM.

2) *ResNet (SSRM Baseline*)*:* Incorporates the SSRM to reshape the 1-D spectral vector into a pseudo-2D tensor for standard 2-D convolutions, but still omits the SA and CA modules.

3) *ResNet + SA:* Incorporates only the SA module into the 2-D ResNet.

4) *ResNet + CA:* Incorporates only the CA module into the 2-D ResNet.

5) *RARNet (Proposed):* The complete architecture that integrates the SSRM and simultaneously utilizes both the SA and CA modules.

The ablation experiment results for SOM and salinity are presented in Table V. The transition from the original ResNet_1D baseline to the ResNet with SSRM yielded a significant improvement in predictive accuracy. In particular, the validation R^2 for the SOM dataset increased by approximately 0.037, while for the salinity dataset, it rose significantly from 0.5247 to 0.6596, marking an improvement of nearly 0.14. These results indicate that introducing the SSRM to reshape the 1-D spectral vector into a pseudo-2D tensor helps 2-D convolutional kernels capture local cross-band interactions more effectively than the 1-D baseline under the current evaluation protocol.

Furthermore, incorporating attention mechanisms into the network demonstrated high efficacy. The results indicate that incorporating either the SA or CA module individually into the SSRM baseline provides notable improvements for specific tasks. For example, on the salinity dataset, the ResNet + CA and ResNet + SA configurations improved the R^2 to 0.7443 and 0.7569, respectively, representing an increase of approximately 0.1 compared to the SSRM baseline without attention mechanisms. However, for the SOM dataset, the addition of either the SA or CA module individually did not enhance the model's performance.

Nevertheless, the proposed RARNet model achieved the highest validation R^2 across both datasets. The R^2 increased to

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TABLE V
ABLATION EXPERIMENT RESULTS FOR SOM AND SALINITY

Soil parameter	Method	Training set			Validation set		
		R^2	RMSE	MAE	R^2	RMSE	MAE
SOM	ResNet_1D (Baseline)	0.9513 ± 0.0468	0.7442 ± 0.5348	0.6130 ± 0.4466	0.6055 ± 0.0591	2.5299 ± 0.3591	2.0265 ± 0.2584
	ResNet (SSRM Baseline)	0.9470 ± 0.0412	0.8492 ± 0.4414	0.6671 ± 0.3499	0.6428 ± 0.0473	2.4096 ± 0.3403	2.0399 ± 0.3055
	ResNet + CA	0.9628 ± 0.0327	0.7116 ± 0.3962	0.5794 ± 0.3187	0.6512 ± 0.0506	2.3658 ± 0.3274	1.9687 ± 0.2761
	ResNet + SA	0.9415 ± 0.0389	0.8824 ± 0.4176	0.6948 ± 0.3315	0.6827 ± 0.0438	2.2483 ± 0.3049	1.8796 ± 0.2518
	RARNet	**0.9120 ± 0.0309**	**1.2075 ± 0.2291**	**0.9782 ± 0.1885**	**0.7291 ± 0.0355**	**2.0986 ± 0.2963**	**1.7449 ± 0.2329**
Salinity	ResNet_1D (Baseline)	0.8761 ± 0.0293	2.3511 ± 0.4185	1.8618 ± 0.3126	0.5247 ± 0.0367	3.9235 ± 2.2793	3.3512 ± 1.9517
	ResNet (SSRM Baseline)	0.9323 ± 0.0174	1.7359 ± 0.2937	1.4018 ± 0.2381	0.6596 ± 0.0938	3.4151 ± 2.3025	2.8925 ± 1.9333
	ResNet + CA	0.9448 ± 0.0149	1.5947 ± 0.2681	1.2945 ± 0.2174	0.7443 ± 0.0746	2.9738 ± 1.9457	2.4526 ± 1.5814
	ResNet + SA	0.9136 ± 0.0188	1.9782 ± 0.3126	1.6027 ± 0.2563	0.7569 ± 0.0625	2.8841 ± 1.7328	2.3479 ± 1.4216
	RARNet	**0.9230 ± 0.0096**	**1.8631 ± 0.2594**	**1.5190 ± 0.2098**	**0.8368 ± 0.0316**	**2.4806 ± 1.5072**	**2.0152 ± 1.2327**

*The results with the highest accuracy have been highlighted in bold. Data are expressed as the mean value ± standard deviation from cross-validation.

TABLE VI
SOM CONTENT CLASSIFICATION STANDARD AND LEVEL DESCRIPTION

SOM level	SOM content (g/kg)	Soil fertility description
Level I	≥40	Highly fertile soil
Level II	30–40	Moderately fertile soil
Level III	20–30	Average fertility soil
Level IV	10–20	Low fertility soil
Level V	6–10	Poor fertility soil
Level VI	<6	Extremely poor fertility soil

TABLE VII
SOIL SALINITY CLASSIFICATION STANDARD AND LEVEL DESCRIPTION

Salinization level	Salt content (g/kg)	Soil salinization description
Level I	< 1.0	Non-salinized soil
Level II	1.0–2.0	Slightly salinized soil
Level III	2.0–4.0	Moderately salinized soil
Level IV	4.0–10.0	Severely salinized soil
Level V	> 10.0	Saline soil

0.7291 for the SOM dataset and 0.8368 for the salinity dataset, representing improvements of over 10% and 20%, respectively. These results demonstrate that the combined use of the SA and CA modules within the SSRM-reconstructed space substantially enhances estimation performance and model stability, outperforming configurations where the modules are applied individually or omitted entirely. This improvement can be explained by the complementary roles of CA and SA. In the SSRM-reconstructed pseudo-2D spectral representation, CA adaptively emphasizes informative feature channels, whereas SA further highlights informative local regions associated with cross-band interaction patterns. Their combination enables RARNet to recalibrate features from both channelwise and local structural perspectives, thereby improving the representation of soil-related spectral information.

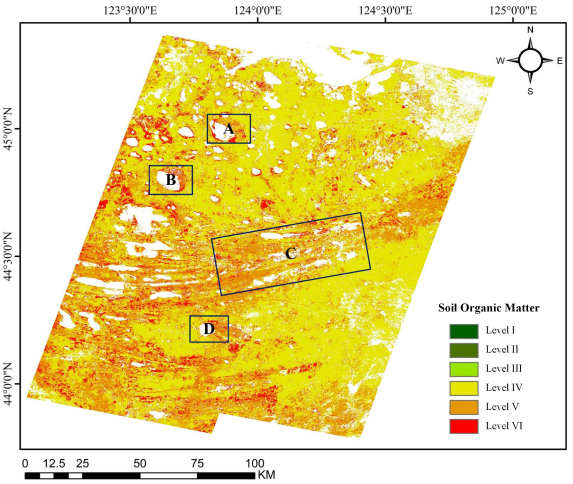


Fig. 9. Classification map for SOM content levels in western Jilin.

D. Mapping Results for SOM and Soil Salinity

The trained RARNet model was applied to satellite hyperspectral imagery to estimate the spatial distribution of SOM and soil salinity. To objectively analyze the distribution characteristics of SOM and salinity in the study area, classification standards were adopted. SOM was categorized into six levels according to the classification scheme of the Second National Soil Survey of China [34], as shown in Table VI. For soil salinity, following the Chinese soil salinization grading standard [35], the degree of soil salinization was divided into five degrees of severity, as detailed in Table VII.

Fig. 9 presents the SOM classification map derived from RARNet. The majority of the study area falls within Levels IV and V, corresponding to low to poor fertility. Levels V and VI are poor and extremely poor soil fertility, and are mainly concentrated around the lake areas (e.g., regions A, B, and D in the image) and the downstream catchment zones within the watershed (e.g., region C), indicating relatively low SOM content in these areas.

Fig. 10 shows the classification map for soil salinity. All five salinity levels are represented in the study area, with the majority of the soils categorized as nonsalinized or slightly salinized. Severely salinized soils and saline soils are primarily concentrated around the lake bodies and reservoirs, including Hua'ao Pao (region A), Dabusuo Pao (region B),

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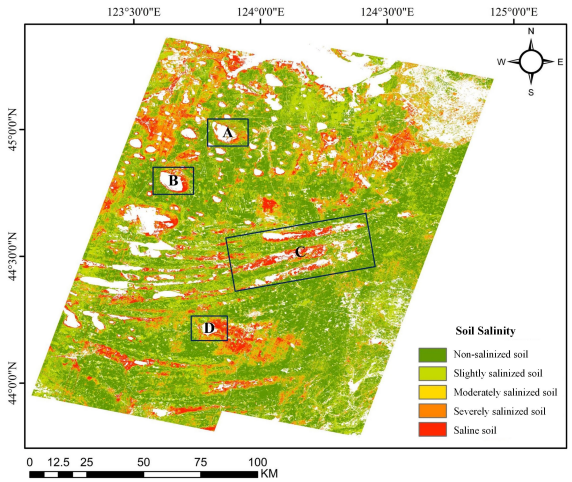


Fig. 10. Classification map for soil salinity content levels in western Jilin.

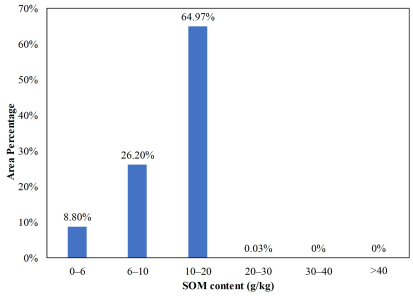


Fig. 11. Distribution of the different SOM content levels.

Chaganhua Pao, and seasonal water accumulation areas in the abandoned river channels (region C), as well as the Longfengshan Reservoir (region D). Distinct strip-like patterns are clearly visible, particularly in the interdune depressions of the longitudinal dunes in Changling County. These strip-like salinization patterns are mainly controlled by lateral river erosion and the orientation of paleo-channels. Low-lying areas along these paleo-channels are prone to water and salt accumulation, resulting in elongated salinization zones.

V. Discussion

A. Analysis of the SOM and Salinity Distribution

Statistical analysis of the estimation results of SOM in the study area reveals the distribution of the different SOM levels, as illustrated in Fig. 11. The proportions of the different SOM levels are as follows: Level III = 0.03%, Level IV = 64.97%, Level V = 26.2%, and Level VI = 8.8%, where the higher numerical grades correspond to lower SOM content. In the study area, the SOM is predominantly classified as Level IV and Level V, indicating that the overall SOM content is relatively low. Previous studies suggest that this low accumulation is likely associated with the relatively enclosed hydrological conditions of the region's lakes and their catchment areas. In particular, the basin topography forms closed-flow lakes where surface water is primarily lost through evaporation [36]. Such hydrological closure reduces the exchange and input of external water bodies, thereby limiting the biological activity

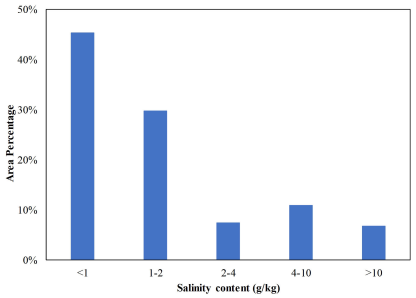


Fig. 12. Distribution of the different soil salinity content levels.

within the water and soil systems and subsequently slowing down the decomposition of organic matter [37]. Furthermore, driven by the region's climate—where annual evaporation (1698 mm) is approximately three times higher than annual precipitation (520 mm)—and intense seasonal freeze-thaw cycles with freezing depths reaching up to 180 cm [38], strong capillary action continuously forces groundwater salts to migrate and accumulate at the soil surface [39]. Meanwhile, intensified hydrodynamic erosion has been reported to further contribute to SOM loss or inhibit its accumulation in similar semiarid topographies [40]. Therefore, these inherent climatic and hydrogeological conditions likely act synergistically to promote salt enrichment while restricting SOM accumulation in these low-lying catchment zones.

Regarding soil salinity, the pixel distribution across salinity levels is displayed in Fig. 12, showing the following proportions: nonsaline soil = 45.3%, slightly saline soil = 29.7%, moderately saline soil = 7.4%, severely saline soil = 10.9%, and saline soil = 6.7%. The generally low SOM content observed across the study region can be related to the widespread salinization, which affects more than half of the total land area. Field surveys confirmed that many former lakebed areas have dried up, with white salt crystals visible on the surface, while signs of waterlogging were also observed in some agricultural zones. These findings suggest that processes such as water overflow and evaporation have led to salt accumulation and subsequent soil salinization. Regions with high salinity levels have largely become bare land, rendering them unsuitable for agriculture.

B. Spatial Relationship Between SOM and Salinity

The spatial distribution of SOM and salinity exhibits a moderately negative correlation across the study area. To further verify this spatial coupling, a bivariate spatial autocorrelation analysis was conducted between the predicted SOM and salinity. The global Moran's I value was -0.2474 ($p < 0.001$), indicating a statistically negative spatial association between SOM and salinity. A bivariate local indicator of spatial association (LISA) cluster map is shown in Fig. 13. The LISA results are dominated by negative spatial correlation clusters. In particular, the widespread high–low clusters, shown in dark green, indicate areas where high SOM values are spatially associated with low salinity levels. More importantly, the low–high clusters, shown in orange, are mainly concentrated along the fringes of lakebeds and banded drainage basins, suggesting that SOM depletion and salinity accumulation are spatially

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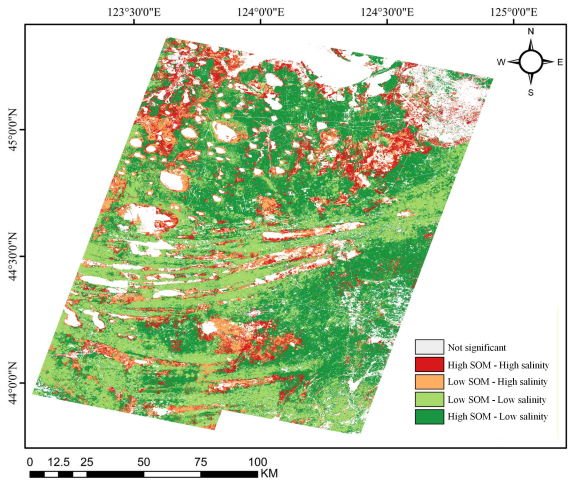


Fig. 13. Bivariate LISA cluster map for SOM and salinity.

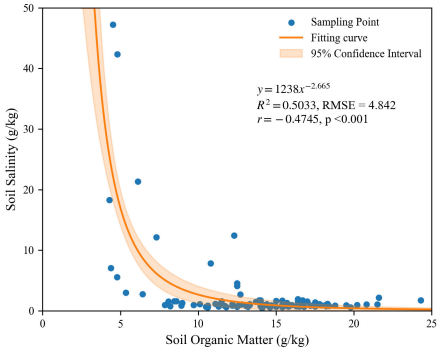


Fig. 14. Scatter plot and corresponding fit curve for SOM and salinity.

coupled in these hydrologically sensitive zones. These areas often feature linear or banded distribution characteristics for both variables, highlighting consistent spatial patterns driven by landscape hydrology and topographic accumulation. Such zones are generally unsuitable for sustainable crop production due to the limited organic content and adverse physicochemical soil conditions.

To quantitatively assess the relationship between SOM and salinity, a regression analysis was conducted using the field sampling data. An exponential regression model was employed, with SOM as the independent variable and salinity as the response. The scatter plot and corresponding fit curve for SOM and salinity are shown in Fig. 14. The model yields an R^2 of 0.5033 and an RMSE of 4.8420. While the RMSE value appears large relative to the mean salinity, this discrepancy is primarily driven by the skewed distribution of salt accumulation in the region. The correlation coefficient between SOM and salinity is −0.4744 with $p < 0.001$, and a covariance of −13.34. These results confirm a moderate but statistically significant inverse correlation and illustrate the inverse spatial pattern, indicating that the depletion of SOM may play a contributing role in the regional salinization process.

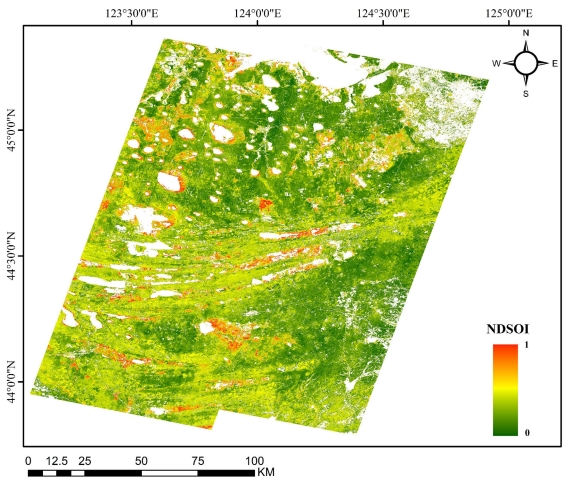


Fig. 15. Distribution map for the NDSOI.

C. Normalized Difference Salt-Organic Matter Index

To better characterize the spatial relationship between SOM and salinity, we propose a new index: the normalized difference salt-organic matter index (NDSOI), as defined in the following equation:

$$NDSOI = norm\left(norm\left(m_{salt}\right) - norm\left(m_{SOM}\right)\right) \tag{4}$$

where $norm(*)$ is the min–max normalization function, and m_{salt} and m_{SOM} denote the salinity and organic matter content, respectively. To ensure the structural robustness of the NDSOI, we conducted a sensitivity analysis using alternative normalization strategies. In particular, the NDSOI was recalculated using Z-score standardization and robust percentile scaling based on the 2nd and 98th percentiles, and the resulting indices were compared with the original min–max-based NDSOI. Both alternative methods produced highly consistent results, with $R^2 = 0.9897$ for Z-score standardization and $R^2 = 0.9801$ for robust percentile scaling. These results suggest that the spatial pattern of the NDSOI is largely insensitive to the choice of normalization method.

This index quantifies the relative contrast between salinity and SOM and ranges from 0 to 1. Lower NDSOI values (closer to 0) indicate a favorable soil environment—i.e., high SOM and low salinity—which is suitable for agricultural production. On the contrary, values approaching one reflect poor soil conditions, with high salinity and low SOM, which are detrimental to plant growth.

The NDSOI was calculated using the SOM and salinity estimation results from the RARNet model. As shown in Fig. 15, high NDSOI values are concentrated around the lakes, reservoirs, and catchment zones. These distributions align with those in Figs. 9 and 10 and were also validated by field observations. To further clarify the agronomic implications of the NDSOI, we linked this index with the regional soil salinization and organic matter classification criteria listed in Tables VI and VII. Specifically, areas simultaneously characterized by severe salinity stress (salinity > 4.0 g/kg) and organic matter depletion (SOM < 10.0 g/kg) were extracted, and their corresponding NDSOI values were statistically ana-

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lyzed. The results showed that the minimum and mean NDSOI values for these severely degraded soils were 0.452 and 0.5478, respectively. Therefore, based on the regional soil classification criteria and the statistical distribution of NDSOI, an empirical threshold of NDSOI > 0.45 was adopted in this study to identify severely degraded soils. This threshold indicates that soils with NDSOI values greater than 0.45 generally exhibit pronounced salinity stress and low organic matter content, indicating potentially degraded soil conditions under the classification criteria used in this study.

We further compared the NDSOI with standalone normalized SOM and normalized salinity using the STD, CV, and Shannon entropy. The STD of the NDSOI was 0.0547, which is lower than those of normalized SOM and normalized salinity, which were 0.0731 and 0.0614, respectively. Its CV was 13.18%, also lower than those of normalized SOM and normalized salinity, which were 31.86% and 137.72%, respectively, indicating that the NDSOI exhibited relatively stable dispersion. The Shannon entropy of the NDSOI was 5.6344, which is higher than that of normalized salinity, which was 3.7685. Although normalized SOM showed the highest entropy value of 6.1713, the NDSOI integrates both SOM and salinity information, providing a more stable and comprehensive representation of their coupled spatial pattern.

The study area is in the southern part of the Songliao Plain. The region is characterized by soda saline-alkali soils dominated by sodium carbonate (Na_2CO_3) and sodium bicarbonate ($NaHCO_3$). The combined effects of elevated salinity and low SOM may deteriorate soil structure, reduce water retention capacity, and weaken soil fertility, thereby constraining local agricultural productivity. To mitigate soil degradation in this region, the cultivation of locally adapted and salt-tolerant crop varieties, including suitable maize and soybean varieties as well as salt-tolerant rice varieties, may serve as a short-term complementary strategy to maintain agricultural utilization under mild salinity stress. Meanwhile, soil improvement in this region should be incorporated into a long-term integrated management framework, including improved drainage and water regulation to reduce salt accumulation, chemical amendments such as gypsum to alleviate soil alkalinity, and continuous organic inputs such as straw returning or organic manure application to replenish SOM and improve soil structure.

VI. Conclusion

In this study, we developed the RARNet model by integrating a deep residual network with an attention mechanism to enhance spatial–spectral feature extraction from hyperspectral data. The model achieved a moderate predictive performance on the validation set in the western Jilin Province, with R^2 values of 0.7291 and 0.8368 for SOM and salinity estimation, respectively. The corresponding RMSE and MAE values were 2.0986 and 1.7449 for SOM, and 2.4806 and 2.0152 for salinity, respectively.

The trained models were applied to multiscene satellite-based hyperspectral imagery to map SOM and salinity across the study area. The results indicate generally low SOM levels and widespread salinization, with more than half of the region affected by varying degrees of salt stress. Areas with SOM below 10.0 g/kg and salinity above 4.0 g/kg were identified as potential high-risk degradation zones with limited fertility and salinity stress. These results may provide preliminary spatial information for future soil management, including drainage regulation, chemical amendments, and organic inputs.

However, several limitations should be acknowledged. The predictive robustness should be interpreted with caution due to the limited number of soil samples and the potential risk of overfitting. Although the SSRM improved model performance in the ablation experiments, its pseudo-2D spectral representation still requires further validation using larger datasets. In addition, the results were derived from a single-season campaign and should be further validated across different soil types, seasons, and wider geographic regions before broader application. The salinity measurement method quantifies total soluble salts but does not distinguish the salt composition or provide electrical conductivity, which may limit its direct agronomic interpretation. Future work will focus on expanding the dataset, incorporating independent validation data, and establishing multiseason collaborative monitoring systems.

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